UMER SIDDIQUI

Senior AI Engineer & Agentic AI Architect | Systems & Automation

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PROFESSIONAL SUMMARY

Senior AI Engineer with deep production experience designing, orchestrating, and deploying real-time Agentic AI systems, event-driven workflow automation (n8n, webhooks), and high-throughput LLM architectures. Extensive track record combining autonomous multi-step reasoning, dynamic tool calling, and contextual memory (LangGraph, Anthropic Claude, OpenAI) with enterprise backend engineering (FastAPI, NestJS, PostgreSQL/pgvector, Redis) and applied computer vision / time-series data modeling. Proven expertise building production telephony voice agents, automated multi-agent scheduling engines, and industrial sensor data analytics pipelines with strict latency, cost, and reliability guarantees.

TECHNICAL SKILLS & STACK

Agentic AI & Orchestration: LangGraph, n8n (Triggers, Webhooks, Branching, Custom Node Logic), CrewAI, Autonomous Tool-Calling, Multi-Agent Planning, Contextual Memory, Human-in-the-Loop Safeguards.

LLMs & APIs: Anthropic Claude (3.5 Sonnet), OpenAI (GPT-4o), Structured JSON Outputs, Dynamic Prompt Engineering, Fallbacks, Retry Mechanisms, API Cost/Latency Optimization.

Backend & Architecture: Python (FastAPI, Flask), Node.js (NestJS), TypeScript, REST APIs, Webhooks, Event-Driven Architecture, Microservices, System Design, DSA.

RAG & Databases: PostgreSQL, pgvector (Hybrid Semantic + SQL Retrieval), Redis Caching, Google Cloud Firestore, SQLite, Embedding Pipelines.

Computer Vision & Data Analysis: PyTorch, YOLO, ResNet (Fine-Tuning), OpenCV, Pandas, NumPy, Time-Series Sensor Analytics, Thermodynamic & Electrical Load Analysis.

Cloud & DevOps: AWS, Google Cloud Platform (GCP), Docker, CI/CD (GitHub Actions), Linux, Postman, Git.

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Saynt AI 06/2025 – Present · Remote

– Architected and deployed production real-time Agentic AI voice workflows with LangGraph and RAG pipelines, featuring webhook-based tool calling to interface with practice-management (PMS) backends (OpenDental, Dentrix) and Twilio telephony with sub-second response times.

– Owned end-to-end backend engineering for the AI voice agent portal and PMS synchronization platform using NestJS/TypeScript and FastAPI, handling concurrent streaming sessions and stateful dialog transitions.

– Engineered robust authentication from scratch, implementing Google OAuth 2.0 and JWT authorization across distributed microservices.

– Designed and automated full CI/CD deployment pipelines on AWS using GitHub Actions, establishing zero-downtime release cadences and structured monitoring.

Jr. AI Developer | Erly Stage Studios 10/2024 – 06/2025 · Remote

– Built the full backend orchestration engine for an autonomous WhatsApp Meeting Scheduling AI Agent in FastAPI, integrating LangGraph with OpenAI and Anthropic models, webhook event handling, and multi-turn state persistence.

– Deployed services on Google Cloud Platform (GCP) with Firestore for session state; collaborated directly with an Anthropic engineer on system scalability, prompt routing, and agent reliability.

– Developed high-throughput product Q&A chatbot API in Flask over 100K+ records in PostgreSQL, combining exact SQL filtering with pgvector similarity embeddings for hybrid structured + semantic retrieval with sub-100ms response times.

AI & Data Analytics Intern | NCL, NED University 04/2024 – 10/2024 · Karachi

– Delivered production-focused AI model development, exploratory data analysis (EDA), and machine learning optimization projects with real-world business constraints.

– Built LocoBot, an autonomous lab assistant robot integrating real-time facial recognition (DeepFace/VGG-Face) with a RAG conversational agent for scheduling faculty meetings, deployed for live student interaction.

KEY TECHNICAL & APPLIED AI PROJECTS

Autonomous Icon CAPTCHA Verification Engine (Computer Vision)

YOLO (Fine-Tuned) · ResNet (Fine-Tuned) · PyTorch · OpenCV · Spatial Geometry

– Engineered an end-to-end autonomous computer vision pipeline solving complex multi-stage icon verification challenges. Fine-tuned YOLO for target symbol detection under heavy visual noise, and fine-tuned ResNet for high-dimensional feature matching and orientation alignment (rotational angles 270°, 225°). Implemented spatial coordinate mapping and automated sequential execution (#1, #2, #3 click targets) with >98% accuracy.

Industrial HVAC VRV Energy & Performance Audit — SGS (British Council Library)

Data Acquisition · Time-Series Analysis · IoT Energy Analyzers · Python · Thermodynamics

– Conducted comprehensive thermodynamic and power performance audits for commercial VRV/VRF HVAC installations for SGS at the British Council Library. Gathered and synchronized continuous high-frequency telemetry across refrigerant temperatures (cooling return, suction, discharge), pressures, and 3-phase power analyzer channels; built Python analytical pipelines to model compressor load cycles, active power fluctuations (18.28 kW to 23.5 kW), and cumulative kWh consumption to deliver actionable energy conservation insights.

Novel Actor-Critic Algorithm for Reinforcement Learning (Final Year Research Project)

Deep Reinforcement Learning · Actor-Critic · PyTorch · OpenAI Gym

– Formulated an architectural modification to the critic value function in standard Actor-Critic architectures, establishing enhanced sample efficiency and convergence stability that surpassed established baselines across multiple benchmark OpenAI Gym environments.

Enterprise Voice AI Telephony Agent

ElevenLabs · Twilio · LangChain · RAG · Practice Management Systems

– Designed and deployed an intelligent voice appointment scheduler with RAG context retrieval, webhook tool-calling, and real-time bidirectional telephony integration, engineered for low-latency production phone calls.

Real-Time Violence Detection System

YOLOv8 · OpenCV · Flask · Model Quantization

– Retrained YOLOv8 on custom datasets for real-time video stream surveillance. Extracted actionable anomaly clips with automated annotated-frame alerting for security operations.

EDUCATION & CREDENTIALS

B.E. Electrical Engineering — NED University of Engineering & Technology · CGPA: 2.6/4.0 · 02/2022 – 08/2026

Higher Secondary (Intermediate) — Bahria College Karsaz · BIEK Board · 996/1100 marks · 2020 – 2022

Secondary School (Matriculation) — CAA Model School and College, Karachi · BSEK Board · 755/850 marks · 2018 – 2020